

NYAYSETU: AN AI-ASSISTED LEGAL SUGGESTION SYSTEM

PROJECT SYNOPSIS
OF PROJECT-I (CSP497)
BACHELOR OF TECHNOLOGY
Computer Science & Engineering

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1. Project Details

- **Problem Statement:** Many individuals cannot access legal assistance due to high consultation costs, complex legal terminology, and lack of awareness. NyaySetu addresses this by using AI to

interpret plain-language descriptions of legal situations, classify them, and guide users toward appropriate professional assistance.

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- **Expected Outcome:** A functional AI-powered legal assistance platform featuring a conversational interface for plain-language input, machine learning-based legal case classification with TF-IDF and ensemble models, and a consultation integration layer connecting users with relevant legal professionals. The system provides informational guidance while maintaining clear ethical boundaries between information and legal advice.

2. Abstract

Access to legal assistance remains out of reach for many individuals due to high costs, limited awareness of legal rights, and the intimidating complexity of legal procedures. Traditional approaches to legal consultation typically demand significant time commitments, substantial financial resources, and physical presence at courts or law offices—barriers that disproportionately affect vulnerable populations.

NyaySetu is a thoughtfully designed system that combines artificial intelligence with practical legal guidance. The system allows users to describe their legal concerns in everyday language, then uses natural language processing (NLP) techniques to understand the query and machine learning algorithms to identify relevant legal categories. Beyond simple classification, the system offers practical suggestions and helps users connect with appropriate legal professionals for further assistance.

By making preliminary legal guidance more accessible while reducing unnecessary consultations, the system addresses genuine needs in legal service delivery. The approach maintains careful ethical boundaries by offering informational support rather than legal advice, ensuring users understand when professional assistance remains necessary. Experimental evaluation confirms that the classification models perform effectively, suggesting the system can scale to serve diverse user populations across different legal contexts.

3. Motivation

The motivation for NyaySetu stems from the severe access inequality in legal services. For countless individuals, obtaining basic legal guidance presents overwhelming obstacles. Legal fees often exceed what ordinary people can afford, legal language remains inaccessible to those without specialized training, and even identifying the right type of lawyer can prove a confusing first hurdle. These difficulties lead many to delay addressing legal matters or navigate complex situations without any guidance, often with serious negative consequences.

Technology has transformed how people access information across nearly every domain, yet legal services have remained surprisingly resistant to such change. The few existing digital legal resources typically assume users already understand their situation well enough to know what questions to ask. NyaySetu was developed to close this gap, meeting users where they are and helping them understand their situations, explore their options, and connect with appropriate assistance.

Furthermore, the economic context makes this work especially important. Students, low-income individuals, and those in rural or underserved areas face compounded barriers. A system that delivers preliminary guidance digitally and routes users to appropriate professionals can meaningfully reduce inequality in access to justice.

4. Literature Review

The intersection of computing and legal practice has drawn increasing research attention over the past decade, with investigators exploring how artificial intelligence might support various aspects of legal work.

Method / Platform	Pros	Cons
Rule-Based Legal Chatbots (Decision Trees)	Predictable and controllable responses within well-defined domains; easy to audit.	Fail when users ask questions outside anticipated patterns; limited flexibility.
Retrieval-Based Legal Q&A Systems	Reliable responses for common inquiries; fast lookup from curated databases.	Fail on novel or complex situations not covered by existing question-answer pairs.
Legal Document Classification (Naive Bayes / SVM)	Computationally efficient; can distinguish broad categories (criminal, civil, family).	Uses formal legal language; underperforms on informal, everyday user queries.
Transformer-Based Models (BERT, Legal-BERT)	Strong contextual understanding; handles nuanced legal language and subtle distinctions.	High computational cost; requires large labelled datasets; complex deployment.
Digital Consultation Platforms (Lawyer Directories)	Connects users directly with practitioners; includes scheduling and video features.	Assumes users already understand their legal situation; no preliminary guidance layer.
NyaySetu (Proposed System)	Integrated plain-language input, ML classification, and consultation matching; ethical information-only stance.	Training data quality challenges; variable performance on highly informal queries.

5. Problem Formulation / Objectives

The challenges preventing people from accessing legal assistance operate at multiple levels. The following objectives guide the NyaySetu system:

- **Bridge the Knowledge Gap (Natural Language Understanding):** Build a conversational interface that accepts plain-language descriptions of legal situations without requiring users to know legal terminology or identify their legal category in advance.
- **Automate Legal Case Classification (Machine Learning):** Develop and train reliable ML classification models (Naive Bayes, SVM, and ensemble approaches) with TF-IDF feature engineering to accurately categorize legal issues and generate probability distributions across legal domains.

- **Deliver Practical Guidance (Information-to-Action Pipeline):** Translate classification outputs into plain-language explanations of likely legal categories, outline available options, and identify situations where professional consultation is strongly recommended.
- **Connect Users with Professionals (Consultation Integration):** Implement a legal practitioner directory filtered by specialization and geography, with scheduling and video consultation features that enable direct connection without requiring users to navigate separate platforms.
- **Maintain Ethical Standards (Information vs. Advice Boundary):** Ensure the system explicitly avoids providing direct legal advice, incorporates clear disclaimers, and uses confidence thresholds to trigger clarifying questions when classifications are uncertain—preserving accuracy and user trust.
- **Ensure Privacy and Accessibility (User-Centric Design):** Collect only information necessary for classification and matching, minimize retention of sensitive data, and ensure the system remains accessible regardless of a user's prior legal knowledge or financial resources.

6. Methodology / Planning of Work

The development of NyaySetu follows a structured, phased process designed to ensure technical reliability, user accessibility, and ethical compliance throughout.

Phase 1: Data Collection & Preparation

- Source training data from public legal databases, anonymized legal aid records, and simulated user queries to cover the full spectrum of potential inputs.
- Perform text preprocessing: tokenization, stop-word removal, normalization of word forms, and filtering of irrelevant characters.
- Ensure balanced coverage across all major legal categories (criminal, civil, family, property, labour, consumer).

Phase 2: Feature Engineering & Model Development

- Apply TF-IDF weighting to identify terms most distinctive across different legal categories.
- Train and evaluate multiple classification algorithms: Naive Bayes (efficiency baseline), Support Vector Machines (stronger per-category performance), and ensemble models (combined reliability).
- Use cross-validation techniques to test model robustness across multiple data partitions.

Phase 3: Conversational Interface Development

- Design a clean, accessible conversational UI that guides users through describing their situation without requiring legal expertise.
- Implement NLP processing layer: tokenization, part-of-speech tagging, and entity recognition tailored to legal contexts.
- Incorporate clarifying prompts that activate when classification confidence falls below defined thresholds.

Phase 4: Guidance Generation & Consultation Integration

- Build the Guidance Generation Layer that translates classification outputs into user-appropriate, plain-language explanations and recommended next steps.

- Implement the Consultation Integration Layer: legal practitioner directory filtered by specialization and geographic availability, with integrated scheduling and video consultation features.
- Apply ethical safeguards: explicit disclaimers, careful phrasing to avoid crossing the information-to-advice boundary, and clear pathways to professional help.

Phase 5: Testing, Evaluation & Deployment

- Evaluate classification accuracy across legal domains with particular focus on informally phrased queries that reflect actual user input.
- Conduct user testing to assess interface clarity, trust, and practical utility.
- Deploy the system and implement data management policies prioritizing user privacy and minimal data retention.

7. Facilities Required for Proposed Work

7.1 Software Requirements

Category	Technology / Tool
Programming Language	Python 3.x
NLP Framework	NLTK, spaCy, scikit-learn
ML / Deep Learning	scikit-learn (SVM, Naive Bayes, Ensemble), Transformers (BERT/Legal-BERT for future extensions)
Feature Engineering	TF-IDF Vectorizer (scikit-learn)
Web Framework (Backend)	Flask or FastAPI (Python)
Frontend / UI	React.js or standard HTML/CSS/JS
Database	PostgreSQL or MongoDB
Version Control	Git, GitHub
Development Environment	VS Code, Jupyter Notebook

7.2 Hardware Requirements

Component	Specification
Development Machine	Standard PC/Laptop with minimum 8 GB RAM, 4-core CPU, 256 GB SSD for development and training.
GPU (optional/cloud)	NVIDIA GPU or Google Colab/AWS SageMaker for training transformer-based models.
Server / Deployment	Cloud instance (AWS EC2 or equivalent) for backend API and consultation matching service.

Internet Connectivity	Stable connection required for API communication, model downloading, and cloud database access.
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